

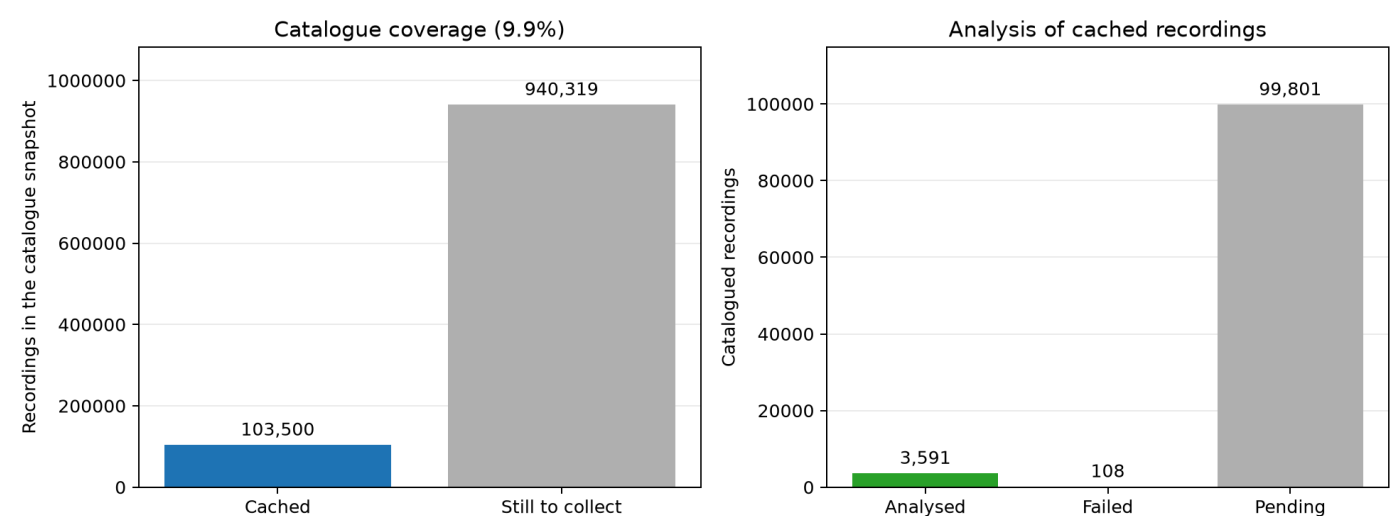
Xeno-canto robustness results

This notebook displays cumulative measurements from the database.

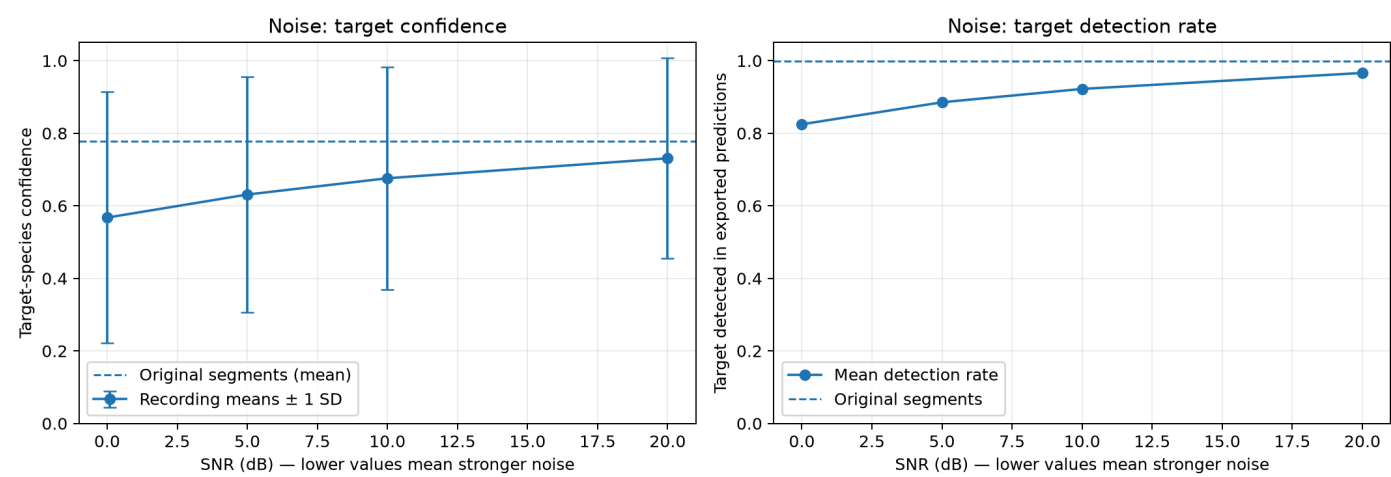
Data and coverage

3,591 recordings analysed · 108 failures · 7 completed batches · cursor 3,699

Corpus progress



Noise robustness



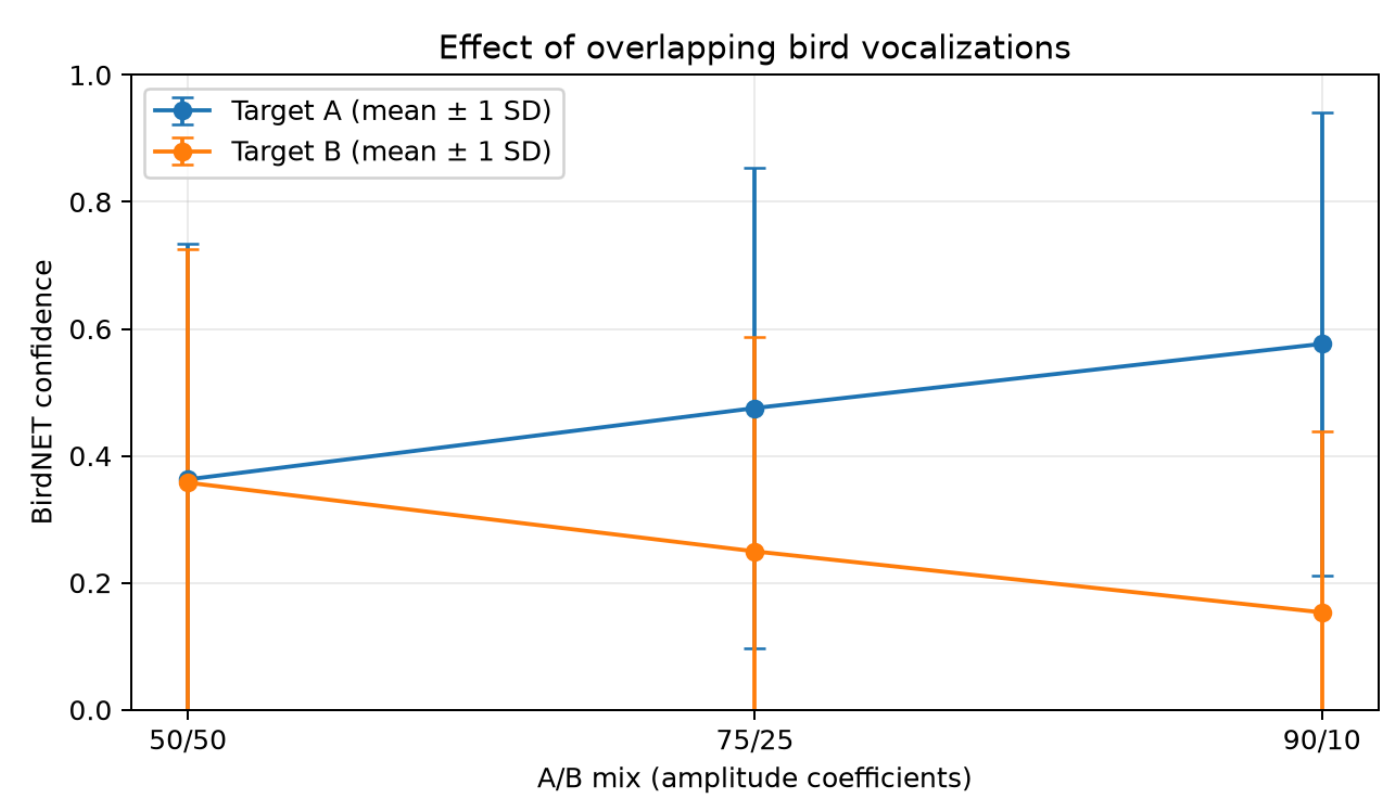
Observation

Segments from **2937 recordings** were tested. Mean target confidence without added noise is **0.778**.

- 20 dB**: mean confidence **0.731**, standard deviation across recordings **0.277**, target detected in **96.6%** of trials.
- 10 dB**: mean confidence **0.676**, standard deviation across recordings **0.307**, target detected in **92.2%** of trials.
- 5 dB**: mean confidence **0.631**, standard deviation across recordings **0.325**, target detected in **88.5%** of trials.
- 0 dB**: mean confidence **0.568**, standard deviation across recordings **0.346**, target detected in **82.5%** of trials.

Error bars show one standard deviation of recording means. A value of 0 means the target is absent from predictions exported at the minimum confidence threshold of 0.10. **654 recordings** without an initially detected target were excluded from this test.

Overlapping vocalizations

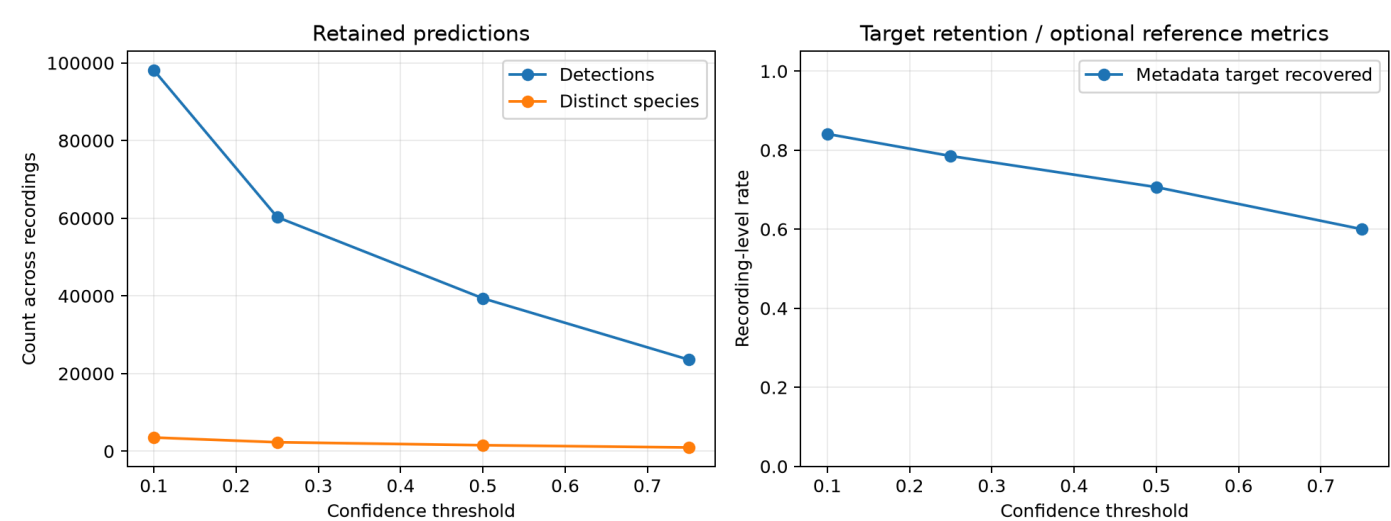


Observation

Means cover **1416 pairs** of distinct recordings. Between mixes 50/50 and 90/10, mean target A confidence changes from **0.363** to **0.576**, and target B confidence from **0.358** to **0.153**.

A and B identify the roles within each pair. Error bars show one standard deviation across pairs. Coefficients apply to amplitude, without prior level equalization.

Confidence threshold



Observation

Between thresholds **0.10** and **0.75**, retained detections change from **98117** to **23583**, and predicted species from **3519** to **960**.

Xeno-canto's primary species is recovered in **84.1%** then **60.0%** of recordings whose target is identified and supported by BirdNET. This rate measures recovery of the known target; recall across all species requires complete annotations.

Precision, recall and F1 are not calculated without complete annotations.

Spectrogram

Example of a short analysed segment.

